

SudoKoders

(Sudoku Board Generator)

SAMPLE INPUT AND OUTPUT

Team members: (C11)

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CASE 1:Valid Move

```
C:\Users\diyan\Documents\SudoKoders>python "Sudoku Generator".py
```

```
Generated puzzle:
```

```
[0, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

```
Type -1 in row to quit and -2 in row to show the full solved puzzle.
```

```
To change the value which you added ,go to the specific cell and then give the desired value or set it to zero to erase the value.
```

```
Enter your move :
```

```
Row (0-8): 0
```

```
Column (0-8): 0
```

```
Number (1-9): 4
```

```
Valid move!
```

```
Updated Puzzle:
```

```
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

```
Enter your move :
```

CASE 2:Invalid Move

Updated Puzzle:

```
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Enter your move :

Row (0-8): 0

Column (0-8): 1

Number (1-9): 5

Invalid move! Puzzle unchanged. Try again...

```
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Row (0-8): 0

Column (0-8): 0

Number (1-9): 0

Case 3:Erasing a previously entered value

Invalid move! Puzzle unchanged. Try again!!!

```
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Row (0-8): 0

Column (0-8): 0

Number (1-9): 0

Valid move!

Updated Puzzle:

```
[0, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Enter your move :

Case 4: Replacing a previously entered value

Updated Puzzle:

```
[0, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Enter your move :

Row (0-8): 0

Column (0-8): 1

Number (1-9): 1

Valid move!

Updated Puzzle:

```
[0, 1, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Row (0-8): 0

Column (0-8): 1

Number (1-9): 4

Valid move!

Updated Puzzle:

```
[0, 4, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Enter your move :

Case 5: Generating a complete solution of the puzzle

Updated Puzzle:

```
[0, 4, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
```

Enter your move :

Row (0-8): -2

Solved Sudoku:

```
[5, 4, 9, 1, 2, 8, 3, 7, 6]
[2, 3, 8, 4, 7, 6, 9, 5, 1]
[1, 7, 6, 3, 5, 9, 8, 4, 2]
[9, 6, 7, 5, 1, 2, 4, 8, 3]
[4, 2, 3, 6, 8, 7, 5, 1, 9]
[8, 5, 1, 9, 4, 3, 6, 2, 7]
[3, 8, 4, 7, 6, 1, 2, 9, 5]
[7, 9, 5, 2, 3, 4, 1, 6, 8]
[6, 1, 2, 8, 9, 5, 7, 3, 4]
```

Enter your next move:

Enter choice 1 to generate a new puzzle:

Enter choice -1 to quit

To change the value which you added ,Change it to zero and then give the desired value.

Enter choice: 1

New Puzzle Generated:

```
[0, 0, 2, 8, 0, 0, 0, 0, 0]
[0, 0, 1, 2, 5, 3, 6, 4, 8]
[0, 3, 4, 0, 0, 0, 0, 0, 0]
[2, 1, 9, 0, 8, 5, 0, 3, 0]
[0, 0, 0, 0, 1, 0, 4, 0, 5]
[4, 6, 0, 9, 0, 7, 0, 8, 1]
[9, 0, 6, 5, 0, 8, 0, 0, 3]
[0, 8, 7, 0, 0, 1, 9, 6, 0]
[0, 0, 0, 7, 6, 9, 0, 5, 4]
```

Case 6: Quitting the game

```
New Puzzle Generated:
[0, 0, 2, 8, 0, 0, 0, 0, 0]
[0, 0, 1, 2, 5, 3, 6, 4, 8]
[0, 3, 4, 0, 0, 0, 0, 0, 0]
[2, 1, 9, 0, 8, 5, 0, 3, 0]
[0, 0, 0, 0, 1, 0, 4, 0, 5]
[4, 6, 0, 9, 0, 7, 0, 8, 1]
[9, 0, 6, 5, 0, 8, 0, 0, 3]
[0, 8, 7, 0, 0, 1, 9, 6, 0]
[0, 0, 0, 7, 6, 9, 0, 5, 4]
Row (0-8): 0
Column (0-8): 0
Number (1-9): 5

Valid move!

Updated Puzzle:
[5, 0, 2, 8, 0, 0, 0, 0, 0]
[0, 0, 1, 2, 5, 3, 6, 4, 8]
[0, 3, 4, 0, 0, 0, 0, 0, 0]
[2, 1, 9, 0, 8, 5, 0, 3, 0]
[0, 0, 0, 0, 1, 0, 4, 0, 5]
[4, 6, 0, 9, 0, 7, 0, 8, 1]
[9, 0, 6, 5, 0, 8, 0, 0, 3]
[0, 8, 7, 0, 0, 1, 9, 6, 0]
[0, 0, 0, 7, 6, 9, 0, 5, 4]

Enter your move :
Row (0-8): -1
You quit the game!
```